

An Exact Schubert and Morse–Bott Atlas for the Grassmann Projection Flow

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Abstract

For a real symmetric matrix A , we solve the Grassmann projection equation $\dot{P} = [P, [P, A]]$ globally by $\text{Ran } P(t) = e^{tA} \text{Ran } P(0)$. Exterior powers give exact Plücker scaling and the basic invariant-subspace limit.

1 Projection dynamics and the exact quotient

Let $A = A^T \in \mathbb{R}^{n \times n}$ and $1 \leq k \leq n - 1$. We realize $\text{Gr}(k, n)$ as the rank- k orthogonal projections

$$\mathcal{P}_{k,n} = \{P : P^T = P, P^2 = P, \text{tr } P = k\}$$

and consider

$$\dot{P} = [P, [P, A]] = AP + PA - 2PAP. \quad (1)$$

The sign in (1) selects the high-eigenvalue subspaces forward in time. The generator A is fixed; the state is the subspace represented by P .

Theorem 1 (Global exact projector). *Let $Q_0 \in \mathbb{R}^{n \times k}$ have full column rank and $\text{Ran } Q_0 = \text{Ran } P_0$. For every $t \in \mathbb{R}$, set*

$$Y(t) = e^{tA} Q_0, \quad P(t) = Y(t)(Y(t)^T Y(t))^{-1} Y(t)^T. \quad (2)$$

Then (2) is the unique global solution of (1); in particular,

$$\text{Ran } P(t) = e^{tA} \text{Ran } P_0.$$

If $Ae_i = \lambda_i e_i$ in an orthonormal eigenbasis, its projective Plücker coordinates obey

$$p_I(t) = e^{t\lambda_I} p_I(0) \quad \text{up to one common scale,} \quad \lambda_I = \sum_{i \in I} \lambda_i. \quad (3)$$

Proof. The matrix e^{tA} is invertible, so $Y(t)$ has full column rank and its Gram matrix is positive definite for all real t . Formula (2) is therefore the orthogonal projector onto $e^{tA} \text{Ran } P_0$. Since $Y' = AY$ and $A = A^T$, differentiation of the Gram inverse gives

$$\dot{P} = AP + PA - 2PAP = [P, [P, A]].$$

Smooth finite-dimensional ODE uniqueness identifies it with the solution. On $\bigwedge^k \mathbb{R}^n$, the induced map sends $e_{i_1} \wedge \cdots \wedge e_{i_k}$ to $e^{t(\lambda_{i_1} + \cdots + \lambda_{i_k})}$ times itself, proving (3). $\square$

This quotient identity is the subspace power-flow mechanism neighboring Oja’s principal-component dynamics and the continuous Grassmann flows of Absil, Sepulchre, and Mahony. We derive every statement needed below and make no priority claim for that classical mechanism.

2 Simple spectrum: every Schubert cell and its rate

Assume in this section

$$\lambda_1 < \lambda_2 < \cdots < \lambda_n, \quad F_j = \text{span}(e_1, \dots, e_j). \quad (4)$$

For a k -plane V , write

$$\mathcal{B}(V) = \{I \subset \{1, \dots, n\} : |I| = k, p_I(V) \neq 0\}.$$

These are the bases of the representable matroid determined by the rows of any frame for V .

Lemma 1 (Matroid guard). *Distinct element weights $\lambda_1, \dots, \lambda_n$ give a unique maximum-weight basis $I_+(V)$ and a unique minimum-weight basis $I_-(V)$ in $\mathcal{B}(V)$, even when two arbitrary k -subset sums coincide.*

Proof. Run the matroid greedy algorithm in decreasing element weight. To see uniqueness directly, suppose two optimal bases differ and take the largest-weight element in their symmetric difference. Strong basis exchange inserts that element into the basis missing it while removing a strictly lower-weight element, producing a heavier basis. This contradiction proves the maximum statement; reverse the weights for the minimum. $\square$

For $I = (i_1 < \cdots < i_k)$, let Ω_I be the Schubert cell characterized by

$$\dim(V \cap F_j) = \#\{a : i_a \leq j\} \quad (1 \leq j \leq n). \quad (5)$$

Equivalently, I is the lexicographically greatest nonzero Plücker index, hence the greedy maximum in Lemma 1.

Theorem 2 (Exact cell limits and actual-support rates). *If $V_0 \in \Omega_I$, then*

$$\lim_{t \rightarrow +\infty} P(t) = P_I,$$

the coordinate projection onto $E_I = \text{span}\{e_i : i \in I\}$. The opposite Schubert cell indexed by $I_-(V_0)$ gives the backward limit.

If $V_0 \neq E_I$, define

$$\rho_2(V_0) = \max_{J \in \mathcal{B}(V_0), J \neq I} \lambda_J, \quad \Delta_+(V_0) = \lambda_I - \rho_2(V_0) > 0. \quad (6)$$

Then

$$\lim_{t \rightarrow +\infty} -\frac{1}{t} \log \|P(t) - P_I\|_F = \Delta_+(V_0). \quad (7)$$

The analogous backward rate uses the actual second-smallest nonzero Plücker weight. For an equilibrium the error is zero and its gap is declared infinite.

Proof. By Lemma 1, I is the unique maximum supported weight despite possible ambient subset-sum ties. Factor $e^{t\lambda_I}$ from the Plücker vector in (3). Every other supported coordinate decays, giving the coordinate-plane limit. The nonzero sum of all weight- ρ_2 coordinate vectors is the first surviving transverse term, so projective distance is $\Theta(e^{-t\Delta_+})$. The Plücker embedding and the orthogonal-projector chart are smooth local embeddings with equivalent norms; the logarithmic rate is therefore (7). Apply the argument to $-A$ for backward time. $\square$

3 Strict Lyapunov law and recurrence

Define $\Phi(P) = \text{tr}(AP)$. Cyclicity of trace and $P^2 = P$ give

$$\frac{d}{dt} \Phi(P(t)) = \text{tr}(A[P, [P, A]]) = \|[P, A]\|_F^2 \geq 0. \quad (8)$$

Equality holds exactly at the invariant projections. If a nonconstant orbit returned arbitrarily close to its initial point, continuity of Φ would force its values back toward $\Phi(P_0)$ after strict increase, contradicting monotonicity. Hence no nonconstant recurrent or periodic trajectory exists. The explicit limits above are stronger than this obstruction.

Source lineage

Oja owns the principal-component learning-flow neighborhood; Brockett owns the double-bracket sorting framework; Absil, Sepulchre, and Mahony directly treat continuous-time subspace flows related to the symmetric eigenproblem. The entries below are ownership citations, not claims of literature priority for this package.

References

- [1] E. Oja, “Simplified neuron model as a principal component analyzer,” *Journal of Mathematical Biology* 15 (1982), 267–273. DOI: [10.1007/BF00275687](https://doi.org/10.1007/BF00275687).
- [2] R. W. Brockett, “Dynamical systems that sort lists, diagonalize matrices, and solve linear programming problems,” *Linear Algebra and its Applications* 146 (1991), 79–91. DOI: [10.1016/0024-3795\(91\)90021-N](https://doi.org/10.1016/0024-3795(91)90021-N).
- [3] P.-A. Absil, R. Sepulchre, and R. Mahony, “Continuous-time subspace flows related to the symmetric eigenproblem,” *Pacific Journal of Optimization* 4 (2008), 179–194. Institutional record: [hdl:2078.5/90452](https://hdl.handle.net/2078.5/90452).